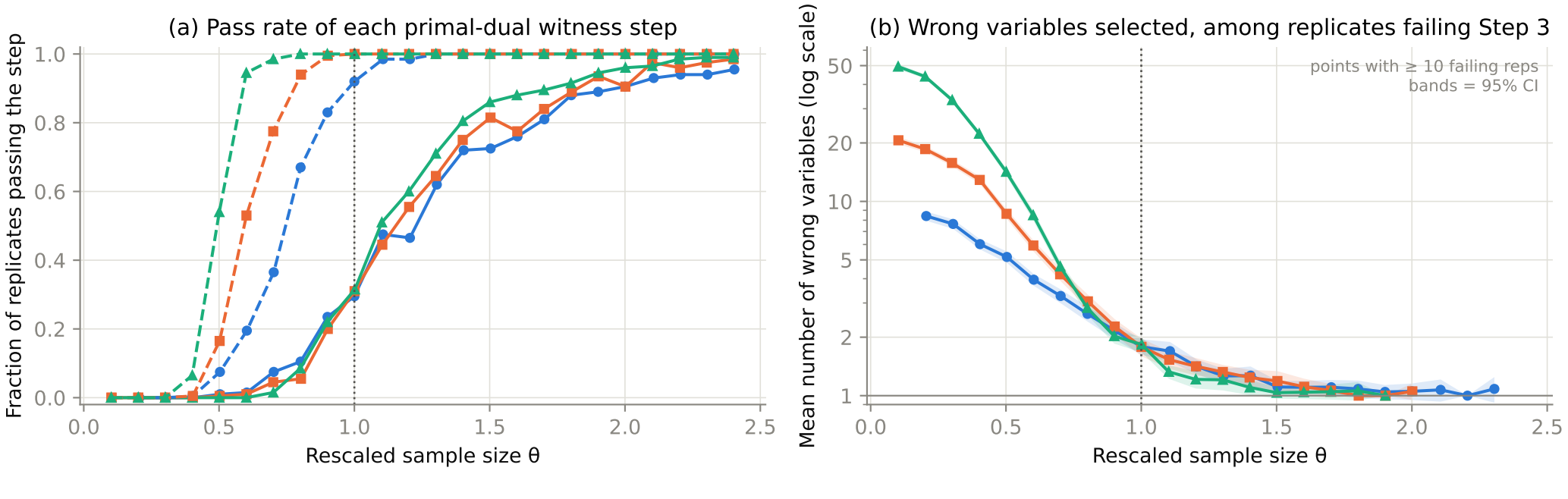
Diagnostics for Fig. 1 (not in the paper): why and how badly the Lasso misses the support
identity design, fractional power sparsity; success requires passing both steps
(a) Pass rate of each primal-dual witness step
Fraction of replicates passing the step
Rescaled sample size θ
(b) Wrong variables selected, among replicates failing Step 3
Mean number of wrong variables (log scale)
points with ≥ 10 failing reps
bands = 95% CI
Rescaled sample size θ
p = 128
p = 256
p = 512
Step 3: no wrong variable enters (dual feasibility)
Step 4: true variables kept with correct signs (sign consistency)